

Surface Cues and a Fine-Tuned Encoder on Directional-Consistent NLI: Baselines for SemEval-2027 Task 2, English Track

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<https://github.com/Fawwad795/dico-nli-semeval2027-group-9>

Abstract

SemEval-2027 Task 2 (DiCo-NLI) asks for one of four relations between two ordered phrases and also scores whether a system’s answers on a pair and on its reversal are compatible. On the English track we ask how much of a fine-tuned encoder’s accuracy and consistency surface cues alone recover. A selective exploratory analysis of 3,042 training instances shows direction leaking through form: the premise is the longer phrase in 78% of forward-entailment instances, and every hypothesis word occurs in the premise in 45% of them. A logistic regression on 17 such cues reaches weighted F1 0.531 and SOFTCONS 0.794 on the development set, the most self-consistent of our classical baselines. DeBERTa-v3-base, fine-tuned over three seeds, reaches 0.807 ± 0.015 weighted F1, 0.877 SOFTCONS and 0.810 HARDCONS. Its advantage sits on instances where length carries no signal and on the boundary between equivalence and entailment; its remaining errors involve one-word substitutions, prepositions and numerals. A constant EQUIVALENCE predictor obtains SOFTCONS 1.0, which is why we read the three official scores only together.

1 Introduction and Task Definition

The task as a black box. DiCo-NLI (DiCo-NLI organizers, 2026) gives a system an ordered pair of short phrases, a premise p and a hypothesis h , and asks for one of four labels, the set $\mathcal{L}$: EQUIVALENCE (the phrases mean the same), FORWARD (p entails h), BACKWARD (h entails p) or NEGATIVE (none of these; Table 1). It is single-label, four-way classification over ordered pairs. Every reversible source pair appears twice, once in each order, and the gold labels of the two orders are tied by a fixed reversal operator: EQUIVALENCE stays EQUIVALENCE, and FORWARD and BACKWARD swap. The organisers report three scores side by side: weighted F1 over all instances, SOFTCONS (do the system’s two answers on a pair obey the

Input	ordered phrase pair (p, h) , English, about three words each
Output	one label per ordered pair
Label space	EQUIVALENCE, FORWARD, BACKWARD, NEGATIVE
Formulation	four-way classification with a cross-instance constraint between (p, h) and (h, p)
Official scores	weighted F1, SOFTCONS, HARDCONS
Our data	track 1 (English): train 3,042, dev 660 instances; dev serves as our test set

Table 1: DiCo-NLI track 1 at a glance.

reversal operator) and HARDCONS (are both answers correct); Section 7 defines them. What a system observes is the two phrase strings and their language codes; the instance and pair identifiers are bookkeeping.

The task as a white box. Whether one phrase entails another depends on whether its meaning is contained in the other’s: in natural logic, adding a restrictive modifier narrows a phrase and yields forward entailment (MacCartney and Manning, 2009). The relevant cues are therefore which phrase carries extra material, whether the heads match, and what quantifiers, numerals and prepositions do to scope. The same structure invites shortcuts: if the more specific phrase is usually the longer one, length and word containment can predict direction without any semantics, and Section 4 shows they do.

Three further confounds matter. The instance identifier leaks the label, since only reversible pairs have a `__flipped` twin, so no model input may include it. A fifth of the development phrases also occur in training. The source of each pair (image captions or news headlines) is not part of the release.

Why the task is hard. The phrases are about three words long and carry no sentence context. Separating equivalence from entailment needs lexical knowledge (“a chocolate bar” and “a candy

bar”), negatives are often a single substituted word inside an identical frame (“off Mexico” and “on Mexico”), and consistency is a property of two predictions that a classifier makes independently.

Question and findings. We ask whether systems reach directional consistency by representing the relation between the two phrases or by exploiting surface asymmetries, and how much of a fine-tuned encoder’s three scores a cue-only model recovers. We test three hypotheses. H1: a cue-only model reaches a large share of the encoder’s SOFTCONS, because its cues are symmetric or change sign exactly under reversal. H2: the encoder’s advantage concentrates where the cues are uninformative, including the boundary between equivalence and entailment. H3: accuracy and consistency trade off in models trained on single ordered instances. H1 and H2 hold; H3 holds inside our classical baselines and fails for the encoder.

2 Related Work

The pilot study. [Apaolaza Larraya et al. \(2026\)](#) define the two consistency measures used by the task as soft and hard causal coherence and evaluate encoder, decoder and encoder-decoder transformers on PhrasIS phrase pairs. Each model was tuned with up to 150 Optuna trials on an 80/20 split of the training data. Encoders lead throughout, with DeBERTa-v3-base reaching weighted F1 0.75, soft coherence 0.84 and hard coherence 0.79 on the combined positives track; decoder-only models trail on both accuracy and coherence, and every family loses ground on unseen pairs. Their manual analysis names prepositional phrases and near-synonym or taxonomic relations as the main sources of error. Their positives track has no negatives and uses a different split, so we read these numbers as a reference point and not as a target.

Direction and the reversal curse. The task takes its framing from [Berglund et al. \(2024\)](#): autoregressive language models fine-tuned on “A is B” fail to answer “B is A”. [Lv et al. \(2024\)](#) trace the failure to the next-token objective and its causal attention, and repair it at training time. DiCo-NLI is an easier setting, since both phrases are given, but the pilot’s decoder results suggest the asymmetry persists. Direction is also fragile in monotonicity reasoning: [Yanaka et al. \(2019\)](#) show that NLI models handle upward inferences far better than downward ones, which is the forward and backward distinction at

	Train	Dev
Instances	3,042	660
Source pairs	1,760	383
reversible (two instances)	1,282	277
negative (one instance)	478	106
Labels E/F/B/N (%)	26/29/29/16	26/29/29/16
Words per phrase, mean (max)	3.0 (15)	2.9 (10)
Missing or empty texts	0	0
Duplicate pairs, groups (rows)	43 (51)	4 (4)
with conflicting labels	1	0
One-word phrases	563	135
Phrases of 8+ words	27	7
Phrases starting with a clitic	56	21
Language pair	en, en	en, en

Table 2: Track 1 splits and quality checks. E/F/B/N: EQUIVALENCE, FORWARD, BACKWARD, NEGATIVE. Duplicate rows are extra copies of an ordered pair; phrase counts are occurrences over both sides of every instance; clitic phrases include those starting with punctuation.

phrase level.

Consistency as a constraint. [Minervini and Riedel \(2018\)](#) and [Li et al. \(2019\)](#) turn logical relations between groups of NLI examples (symmetry of contradiction, transitivity) into training penalties and improve both accuracy and consistency. [Elazar et al. \(2021\)](#) define consistency as invariance under meaning-preserving rewording and measure it on ParaRel. Reversal is a constraint of the same family, with the difference that the label must change in a prescribed way.

Data and models. The phrases come from PhrasIS ([Lopez-Gazpio et al., 2024](#)), which annotates phrase pairs extracted from the chunk alignments of interpretable semantic textual similarity ([Agirre et al., 2016](#)); this origin explains why items are noun phrases, verb chains and prepositional phrases. Our encoder baseline is DeBERTa-v3-base ([He et al., 2023](#)), the pilot’s strongest base-size model.

3 Dataset

We use the track 1 (English) training and development files of the official release at commit 588968e (8 September 2026). Each reference row holds an instance identifier, a source-pair identifier, the two phrases with language codes, the identifier of its reversed twin and the gold label. Table 2 gives the split statistics and the quality checks.

Splits. The official test set appears only on 10 January 2027, after the course ends, so the devel-

Group	Question
Length	Does the longer phrase predict the direction?
Readability	Do word and syllable lengths differ by relation?
Lexical	Do overlap and containment separate direction, equivalence and negatives?
Semantic	Does embedding similarity separate the relations, and can it see direction?
Linguistic	Do heads, POS patterns, adjectives and prepositions vary by relation?
Task cues	Are quantifiers, negation, numerals and named entities frequent enough to matter?

Table 3: The six EDA feature groups and the question each answers.

opment set is our test set. We never select anything on it: model selection uses slices of the training data split by source pair, so that both directions of a pair stay on the same side. Train and dev share no source pair, but 7 unordered phrase pairs and 155 of the 712 distinct development phrases (22%) also occur in training.

Quality. The data is complete, and the one real defect is small: “South Africans” and “South Africa” appear in training once as FORWARD and once as NEGATIVE. One-word phrases are common, and phrases such as “’s war shrine visit” are fragments of the source chunking. The image-caption versus news-headline origin is absent from the train and dev files; only the 461-pair trial sample records it (244 caption and 217 headline pairs), so we cannot stratify by source.

4 Exploratory Data Analysis

We analyse the training split only and choose six feature groups, each to answer one question about the task (Table 3). Lexical-richness measures such as TTR, MTLT and HD-D are left out: on phrases of about three words they are undefined or unstable. All seven figures of the analysis are collected, uncaptioned, in the separate EDA PDF; two appear below.

Length. *Observation:* the premise is longer in 78% of FORWARD instances and the hypothesis in 78% of BACKWARD instances, while EQUIVALENCE splits evenly (36% premise longer, 29% equal, 36% hypothesis longer) and half of the NEGATIVE instances (49%) have equal length (Figure 1). *Interpretation:* direction leaks through form, because the entailing phrase is usually the more specific one with an added modifier; this is a shortcut

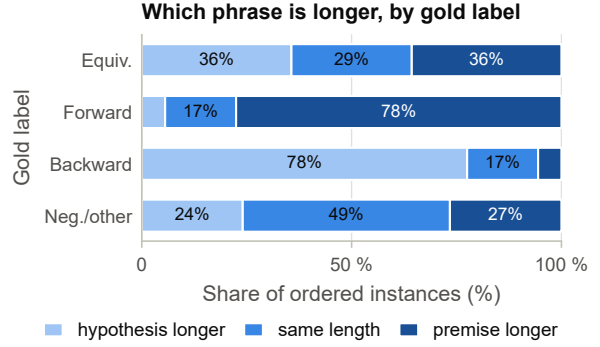


Figure 1: Which phrase is longer, by gold label (train).

a model can learn without semantics. *Implication:* length belongs in the cue baseline, and every model should also be scored on instances where length points the wrong way or nowhere (Section 9).

Lexical overlap and containment. *Observation:* mean word-set Jaccard overlap is 0.41 for directional pairs against 0.24 for EQUIVALENCE and 0.27 for NEGATIVE. Every hypothesis word occurs in the premise in 45% of FORWARD instances and 2% of BACKWARD instances, and the mirror holds; the last word is shared in 58% of directional instances (Figure 2). *Interpretation:* directional pairs are mostly one phrase with words added or removed around a kept head, while equivalences are paraphrases with little literal overlap. *Implication:* containment in both orientations is a strong but partial direction cue, firing on fewer than half of the directional pairs; overlap signals relatedness, not equivalence.

Semantic similarity. *Observation:* cosine similarity of MiniLM sentence embeddings (Reimers and Gurevych, 2019) has median 0.76 for EQUIVALENCE, 0.77 for both directions and 0.64 for NEGATIVE; 52% of directional and 50% of EQUIVALENCE instances lie at or above the EQUIVALENCE median, against 27% of negatives. Cosine is symmetric, so a pair and its reversal get the same value. *Interpretation:* similarity separates negatives from related pairs moderately and EQUIVALENCE from entailment not at all, and it carries no information about direction. *Implication:* a symmetric representation cannot recover direction, so we leave cosine out of the cue set and expect the encoder, which reads the ordered pair, to be needed for the equivalence boundary.

Linguistic and task-specific cues. *Observation:* the two heads share a lemma in 52% of directional

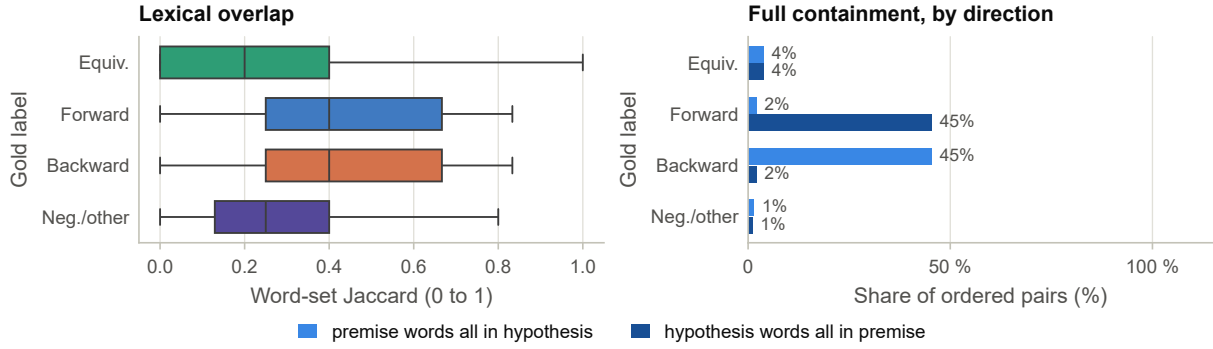


Figure 2: Word overlap (left) and full containment in each direction (right), by gold label (train).

instances against 30% of EQUIVALENCE; the part-of-speech pattern is identical in 31% of NEGATIVE instances against 7% of directional ones; the premise of a FORWARD pair carries 0.33 more adjectives on average. Numerals occur in 18% of directional instances, named entities in 47%, while quantifiers (3%) and negation (under 1%) are rare. *Interpretation:* negatives are typically a single substitution inside an unchanged frame, and adjectives are the material that narrows a phrase. *Implication:* head and modifier comparisons matter more than global similarity; quantifiers and negation are too rare to train on and serve as error-analysis slices instead.

Readability. *Observation:* syllables per word range from 1.53 to 1.60 and characters per word from 4.5 to 4.9 across labels, and median Flesch reading ease from 77.9 to 82.4. *Interpretation:* the phrases are uniformly short caption and headline language; Flesch assumes sentences and here tracks syllable density only. *Implication:* readability adds nothing to separate the labels and stays out of the feature set.

5 Feature Engineering

The classical baselines use two feature sets, both computed from the two phrase strings only; a unit test checks that no identifier reaches the feature matrix.

Surface cues. Seventeen numeric cues turn the EDA findings into features (Table 4). Each cue is either unchanged when the pair is reversed (overlap, shared words) or changes sign or swaps with its partner (length differences, the two containments), which matters for consistency in Section 8. Readability and embedding similarity are excluded for the reasons given in Section 4. The cues are stan-

Group	Cues
Length (7)	words and characters per side, their differences, log word ratio
Overlap (2)	shared words, Jaccard
Containment (4)	share of each side’s words in the other; each side contained as a contiguous span
Position (2)	same first word, same last word
Numerals (2)	a digit on each side

Table 4: The 17 surface cues, grouped.

dardised on the training data.

Lexical n-grams. TF-IDF over word unigrams and bigrams of the string “ p [SEP] h ”, with sub-linear term frequency. The token pattern keeps one-letter words and the separator as a token of its own, so bigrams that cross the separator encode which phrase a word belongs to.

6 Baseline Models

All settings below were fixed before the first run and none was tuned on the development set.

Trivial systems. A majority-label predictor, a uniform random predictor (seed 13) and two constant predictors, EQUIVALENCE everywhere and NEGATIVE everywhere. Training ties FORWARD and BACKWARD at 881 instances each; the tie goes to the first label in canonical order, so the majority system predicts FORWARD everywhere. The constant systems probe the metrics (Section 7).

Classical baselines. Logistic regression on the cues, on the n-grams, and on both side by side, and gradient-boosted trees on the cues, all in scikit-learn (Pedregosa et al., 2011) with balanced class weights; the regressions use $C = 1$, the trees 300 iterations at learning rate 0.05, and every other setting is a library default.

Pretrained baseline. DeBERTa-v3-base (He et al., 2023) with a classification head, fully fine-tuned with Hugging Face Transformers (Wolf et al., 2020) in our own training loop. The pair is tokenised as one sequence of at most 32 tokens. We train for 5 epochs with AdamW, learning rate 2×10^{-5} , batch size 16, weight decay 0.01, a linear schedule with 6% warm-up and gradient clipping at 1.0, and keep the epoch with the best weighted F1 on a 20% slice of the training data split by source pair. We train three seeds (13, 42, 1234); each seed also draws its own slice. Each run trained in about 140 seconds on one NVIDIA T4, and a second execution reproduced every development prediction exactly.

7 Evaluation Setup

Weighted F1. With n_c gold instances of label c among N and F_c the F1 of label c ,

$$\text{wF1} = \sum_{c \in \mathcal{L}} \frac{n_c}{N} F_c, \quad F_c = \frac{2P_c R_c}{P_c + R_c}. \quad (1)$$

Each label counts in proportion to its frequency, so NEGATIVE, 16% of the data and the hardest class for every model, weighs least. Macro F1 averages F_c with equal weights, and micro F1 equals accuracy for single-label prediction; we report macro F1 next to the official score so that a weak minority class stays visible.

Consistency. Let Rev map EQUIVALENCE to EQUIVALENCE, FORWARD to BACKWARD and BACKWARD to FORWARD; NEGATIVE has no reverse. Let $\mathcal{P}$ be the set of reversible pairs (x, x^r) in the development set ($|\mathcal{P}| = 277$), f the system, y the gold label and $\mathbb{I}[\cdot]$ the indicator:

$$\text{SOFTCONS} = \frac{1}{|\mathcal{P}|} \sum_{\mathcal{P}} \mathbb{I}[f(x) = \text{Rev}(f(x^r))], \quad (2)$$

$$\text{HARDCONS} = \frac{1}{|\mathcal{P}|} \sum_{\mathcal{P}} \mathbb{I}[f(x) = y \wedge f(x^r) = y^r]. \quad (3)$$

Because gold labels obey the reversal, a pair that is right in both directions is also consistent, so $\text{HARDCONS} \leq \text{SOFTCONS}$. Negative pairs have no reversed twin and enter only weighted F1; a NEGATIVE prediction on a reversible pair counts as incompatible.

What each score rewards. Weighted F1 rewards correct labels on single instances, SOFTCONS rewards self-consistency whether or not it is correct, and HARDCONS rewards correctness on whole pairs. The constant systems show why the task

reports all three. Predicting EQUIVALENCE everywhere is consistent on every pair, so it scores SOFTCONS 1.000 with weighted F1 0.110, and its HARDCONS of 0.314 is the share of reversible pairs whose gold label is EQUIVALENCE (87 of 277). Predicting NEGATIVE everywhere scores 0.044, 0 and 0.

Protocol. Every score is computed by the official scorer, run unchanged as the competition platform runs it. The classical baselines are also cross-validated on the training data in five folds grouped by source pair, every fold scored by the same scorer. Each run records its seed, the data commit, the code commit and the exact command; Appendix A maps every number in this report to its run.

8 Results

Table 5 lists every system on the development set. Cross-validation on the training data ranks the classical systems in the same order as the development set. Every development score is at or below its cross-validated mean, by 0.1 to 1.2 standard deviations and by 2.0 for n-grams alone, so the development set is somewhat harder than held-out training folds.

Surface cues buy consistency (H1). The cue-only model is the most self-consistent classical system: 220 of the 277 reversible pairs get compatible answers, and it confuses FORWARD with BACKWARD in only 8 of the 380 directional instances. Its weights follow the EDA (Table 6): hypothesis-in-premise containment pushes towards FORWARD and away from BACKWARD, its mirror does the reverse, and the length differences add a smaller push of the same kind. Because these cues swap or change sign under reversal, the model’s decisions on a pair and its reversal are compatible almost by construction. Consistency is not correctness, though: 67 of its 220 consistent pairs are wrong in both directions, so HARDCONS stays at 0.552.

Inside the classical family, accuracy costs consistency (H3). Adding n-grams to the cues raises weighted F1 from 0.531 to 0.596 and lowers SOFTCONS from 0.794 to 0.747; trees on the same cues show the same trade (0.574 and 0.718). The more expressive models find order-specific patterns that help single instances and break agreement across the pair. N-grams alone see the order only through bigrams that cross the separator, and score lowest on every metric.

System	Weighted F1	Macro F1	SOFTCONS	HARDCONS	CV weighted F1
Majority (FORWARD everywhere)	0.129	0.112	0.000	0.000	
Random, seed 13	0.246	0.240	0.184	0.054	
EQUIVALENCE everywhere	0.110	0.104	1.000	0.314	
NEGATIVE everywhere	0.044	0.069	0.000	0.000	
Cues, logistic regression	0.531	0.502	0.794	0.552	0.575 ± 0.037
N-grams, logistic regression	0.407	0.399	0.426	0.256	0.429 ± 0.011
Cues and n-grams, logistic regression	0.596	0.573	0.747	0.570	0.617 ± 0.021
Cues, gradient-boosted trees	0.574	0.550	0.718	0.527	0.572 ± 0.024
DeBERTa-v3-base, mean $\pm$ sd of 3 seeds	0.807 ± 0.015	0.786 ± 0.017	0.877 ± 0.010	0.810 ± 0.013	
Pilot DeBERTa-v3-base, other track [†]	0.75	n/a	0.84	0.79	

Table 5: Track 1 development set, official scorer. CV: cross-validation on the training data, five folds grouped by source pair, mean $\pm$ sd. For DeBERTa the held-out weighted F1 at the selected epoch averages 0.798. [†]Reported by [Apaolaza Larraya et al. \(2026\)](#) on the PhrasIS positives track, without negatives and on another split; a reference point, not comparable.

Cue	E	F	B	N
h words in p	-0.09	+0.96	-1.44	+0.57
p words in h	-0.02	-1.35	+0.97	+0.41
Jaccard overlap	-0.42	+0.84	+0.82	-1.25
Character difference	+0.01	+0.49	-0.49	-0.01
Log word ratio	-0.02	+0.39	-0.38	+0.02

Table 6: The five largest standardised weights of the cue-only logistic regression, per label (E/F/B/N: EQUIVALENCE, FORWARD, BACKWARD, NEGATIVE).

The encoder gains on all three (H3 fails for it). DeBERTa-v3-base raises weighted F1 by 0.21 and HARDCONS by 0.24 over the best classical system, and SOFTCONS by 0.08 over the most consistent one (Figure 3). The gain is largest where the cues were weakest: against the combined classical model, F1 on EQUIVALENCE rises from 0.45 to 0.78 and on NEGATIVE from 0.41 to 0.63, while the two directions rise from 0.72 and 0.71 to 0.86 and 0.87. Its consistency follows from correctness: only 16 to 21 of its consistent pairs per seed are wrong. We do not claim it beats the pilot, whose numbers come from another track, another split and a 150-trial search.

9 Error Analysis

How pairs fail. Table 7 sorts each reversible development pair by what a system did with its two directions. The cue model almost never gets one direction right and the other wrong (1 pair): its errors are symmetric, either consistent and wrong (67) or wrong both ways (56). Adding n-grams turns 21 pairs into half-right ones, the order-specific behaviour behind its lower SOFTCONS. DeBERTa leaves 49 to 56 pairs not fully correct per seed,

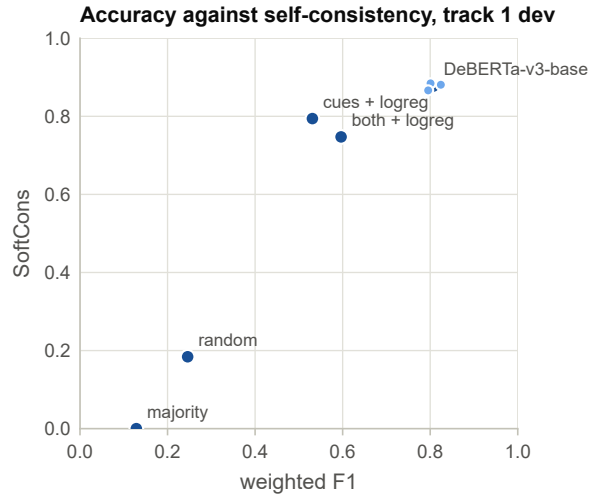


Figure 3: Weighted F1 against SOFTCONS on the development set; small points are the three DeBERTa seeds.

spread evenly over the three failure types.

Where the encoder wins (H2). We split the development instances by what the length cue says about the gold label (Table 8). It agrees on 282 instances, points the wrong way on 10, and says nothing on 368 (equal length, or a label without direction). Where it agrees, the cue model is already strong and the encoder adds 0.02 weighted F1. Where it says nothing, the cue model falls to 0.31 and the encoder holds 0.75, a gap of 0.44.

The misleading slice is too small to read. Slice scores are classification scores only, since a slice can separate the two directions of a pair.

What the encoder still gets wrong. DeBERTa rarely confuses direction: seed 13 swaps FORWARD and BACKWARD on 3 instances. Its errors sit on the EQUIVALENCE and NEGATIVE bound-

System	Consistent		Inconsistent	
	right	wrong	one right	both wrong
Cues	153	67	1	56
Cues and n-grams	158	49	21	49
DeBERTa, seed 13	228	16	15	18
DeBERTa, seed 42	224	21	12	20
DeBERTa, seed 1234	221	19	17	20

Table 7: Outcomes on the 277 reversible development pairs. A consistent pair is either right in both directions or wrong in both.

Length cue	<i>n</i>	Cues	Cues + n-grams	DeBERTa
Agrees	282	0.907	0.917	0.931
Misleads	10	0.000	0.000	0.800
Uninformative	368	0.313	0.425	0.754

Table 8: Weighted F1 on development slices defined by the length cue; DeBERTa is the mean of three seeds.

aries: 26 directional instances predicted as EQUIVALENCE, 19 EQUIVALENCE instances predicted as NEGATIVE, and 34 of 106 negatives called related. All three seeds miss the same 80 instances, and the cue model misses 50 of those too. We read 20 hand-picked errors of seed 13 and sorted them into five categories modelled on the pilot’s manual analysis (Table 9); the counts are four per category by construction, so they illustrate the kinds of error and do not estimate their rates.

Some errors may be in the gold labels. “two people” and “2” are labelled NEGATIVE, and “Saudi Women” to “Saudis” is labelled BACKWARD although the first phrase is the more specific one; with the conflicting “South Africans” duplicate in training (Section 3), this suggests a share of the remaining errors are annotation disagreements. We have not audited the labels systematically.

10 Discussion

Direction is the easy part. Seventeen surface cues decide FORWARD against BACKWARD almost as reliably as a fine-tuned encoder (8 against 3 confusions out of 380). What the cues cannot decide is the kind of relation: whether added material narrows the meaning or paraphrases it, and whether a substituted word breaks the relation. That is where most of the encoder’s 0.21 gain in weighted F1 comes from (Table 8), and where its remaining errors are.

SOFTCONS on its own is easy to game. A constant EQUIVALENCE predictor scores a perfect

1.000, and the cue model reaches 0.794 while being wrong in both directions on a quarter of the pairs. A system can be consistent by construction, through symmetric or sign-flipping inputs, without understanding the relation. We therefore read SOFTCONS only next to HARDCONS and macro F1, and count consistent-but-wrong pairs explicitly.

The encoder is consistent because it is correct.

The accuracy-consistency trade within the classical family did not carry over: DeBERTa improved all three scores at once, and its consistent pairs are almost all correct ones. This makes the open question for decoder models, which the pilot found far less coherent, sharper: their consistency gap may be a correctness gap on the hard relations and not a separate defect.

How far the numbers travel. The development set behaves a little harder than held-out training folds, a fifth of its phrases also occur in training, and some gold labels look questionable. Differences of a few points between systems should be read with these in mind; the differences we draw conclusions from are 0.2 or more.

11 Next Steps

Assignment 2 moves to decoder language models, where the pilot found the largest consistency gap. We will compare open-weight instruction-tuned models used by prompting and fine-tuned with low-rank adapters, and test three mechanisms for direction: training on both orders of every pair, a loss term that penalises incompatible predictions on a pair and its reversal in the manner of Li et al. (2019), and prompts that present both orders together. Every system will be reported with weighted F1, macro F1, SOFTCONS, HARDCONS, the consistent-but-wrong count and the length-cue slices of Table 8, against the baselines of this report. We will also audit the conflicting and questionable gold labels before relying on the error categories, and add the Spanish and Basque tracks if time allows.

Limitations

All results are on one track, English, and on the development set, which we use as a test set because the official test set is not yet released. The classical baselines use one seed and fixed settings, and the encoder three seeds with a fixed recipe; neither was tuned, so the numbers are baselines and not upper

Category	Premise	Hypothesis	Gold	Predicted
Preposition	at a pool	in front of a pool	BACKWARD	NEGATIVE
Numeral or quantifier	A group of people	Six people	FORWARD	EQUIVALENCE
Near-synonym or taxonomy	a chocolate bar	a candy bar	FORWARD	EQUIVALENCE
One-word negative contrast	on a crowded street	on a public street	NEGATIVE	EQUIVALENCE
Chunking fragment	in destroying	destroy	FORWARD	EQUIVALENCE

Table 9: One example per error category, DeBERTa seed 13 on the development set.

bounds. The misleading length-cue slice has 10 instances. The error categories come from 20 hand-picked errors and describe kinds of failure, not their frequency. The part-of-speech and similarity features rely on spaCy’s small English model and on one sentence encoder, both applied to phrases without context.

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A Where Every Number Comes From

Each run folder under experiments/ holds a config.json with the seed, the data commit (588968e), the code commit and the exact command, together with the official scorer’s report for every system. The notebooks under notebooks/assignment-1/ rerun headlessly with `uv run jupyter execute`; the encoder was trained by the Modal launcher `scripts/modal/train_encoder.py` and its outputs fetched before scoring.

- Data checks and EDA: notebook 02_eda_track1, folder 2026-09-29-eda-track1.
- Trivial and constant systems: notebook 01_trivial_baselines, folder 2026-09-29-trivial-baselines-track1.
- Classical baselines: notebook 03_ml_baseline_track1, folder 2026-09-29-ml-baseline-track1.
- DeBERTa-v3-base: notebook 04_dl_baseline_track1, folder 2026-09-29-deberta-base-track1.
- Error analysis: notebook 05_error_analysis_track1, folder 2026-10-02-error-analysis-track1.